

Voss Lundberg

The tornado chart was the first slide in the appendix and the last thing anyone looked at, which was unfortunate, because it contained the finding.

Voss Lundberg was a Nordic industrial software company that the Ravensbourne Partners deal team had been working for four months. Revenue of 1,400 million kronor, an operating margin of 15.5 percent, a cost of capital of 8.8 percent, and terminal growth of 2.8 percent produced an enterprise value of about 2,011 million. The team's recommendation was to bid.

The associate on the deal, Tomás Iriarte, had built a Monte Carlo around the model at the partner's request. It carried fourteen stochastic inputs: revenue growth for each of the five forecast years, operating margin for each of the five years, the reinvestment rate, working capital intensity, the tax rate, and terminal growth. Each had a triangular distribution with a low, a base, and a high supplied by the sector team. Ten thousand draws produced a distribution of enterprise value with a mean of 2,040 million, a fifth percentile of 1,510, and a ninety-fifth of 2,690. The partner had found the range reassuring.

The cost of capital was not one of the fourteen. It sat in a single cell at 8.8 percent, computed from a beta of 1.15, a risk-free rate of 2.9 percent, an equity risk premium of 5.0 percent, and a target capital structure of 25 percent debt. Iriarte had not made it stochastic because the sector team had not given him a range for it, and because, as he put it in the file note, "the WACC is a policy input rather than a forecast."

Marja Eklund, the operating partner assigned to the deal, had spent eleven years running a business that competed with Voss Lundberg. She read the tornado chart in the appendix and asked one question in the Monday meeting.

"You have moved every line in the P&L and you have not moved the discount rate. Why is the discount rate the one number we know?"

There was a second thing in the appendix that Eklund had not asked about, and that Iriarte had been quietly worried about since he built it. Roughly three-quarters of the enterprise value sat in the terminal value, and terminal growth was one of the fourteen stochastic inputs — drawn independently of every other input, including the margin path that produced the cash flow the terminal value capitalized. In the draws where margins came out at the low end, terminal growth was just as likely to come out at the high end, which described no company Eklund had ever competed against.

EXHIBIT 1 — BASE CASE AND ONE-AT-A-TIME SENSITIVITY

Enterprise value at the base case: 2,011 million kronor. Terminal value is 76.5 percent of it.

INPUT	RANGE TESTED	EV AT LOW	EV AT HIGH	SWING
Revenue growth	4% to 8%	1,846	2,189	343
Operating margin	13.5% to 17.5%	1,751	2,270	519
Terminal growth	1.8% to 3.8%	1,778	2,336	558
Cost of capital	7.8% to 9.8%	2,419	1,720	699

The cost of capital row was computed by the sector team for the appendix. It was not included in the simulation.

ASSIGNMENT

- 1 Rank the four inputs by the size of their effect on enterprise value. Which one was left deterministic, and where does it rank?
- 2 Iriarte wrote that the cost of capital is a policy input rather than a forecast. Is that a good reason to hold it fixed in a simulation? Answer in terms of what the simulation is for.
- 3 Add the cost of capital to the simulation with a plausible distribution. State the distribution and justify it, then report what happens to the fifth percentile of enterprise value.
- 4 Terminal growth and the margin path are drawn independently. Construct a case in which that is defensible and a case in which it is not. Then impose a dependence you can defend and report the effect.
- 5 Voss Lundberg's cash flows are discounted at a risk-adjusted 8.8 percent and then simulated. Is the risk being counted twice? Answer carefully — the answer is not simply yes or no.